CSE221: Algorithms

Graphs

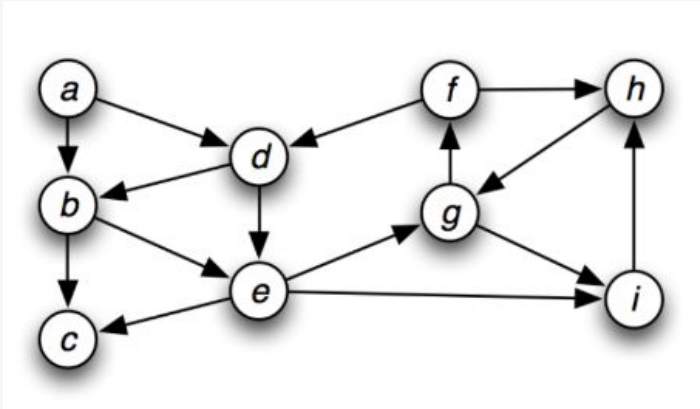
Prepared by: A B M Muntasir Rahman [MUNR]



What is a Graph?

- Graph G is pair (V, E) where V is a finite set (set of vertices) and E is a finite set of pairs from V (set of edges).
- We generally denote n = |V| and m = |E|.

Here,
V = {a, b, …, i} and n = |V| = 9
E = {ab, bc, ad, …..} and m = |E| = 15

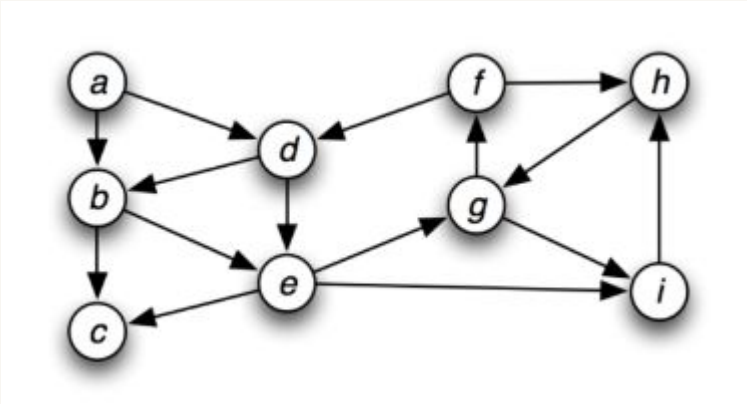


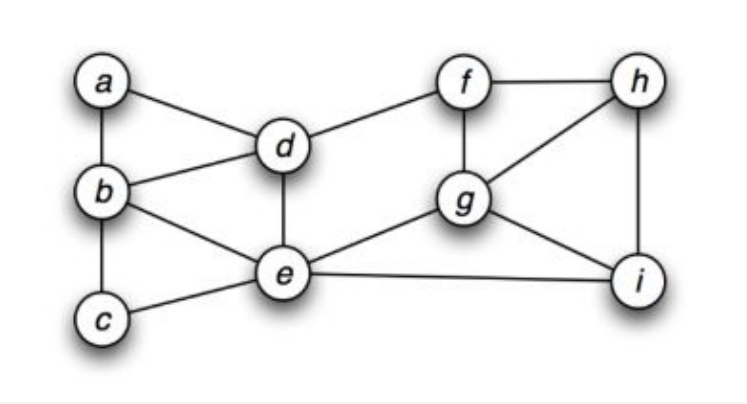
Applications of Graphs

- **Transportation Networks:** How should you design the roads and highway network in a country? What is the quickest way to go from BRAC University to Mirpur?
- **Communication Networks:** How to send network packet from your computer to your friend's computer?
- **Social Networks:** Facebook, Instagram, Twitter etc.
- **Dependency Networks:** What courses you must take before you can take CSE470?

Directed vs Undirected Graphs

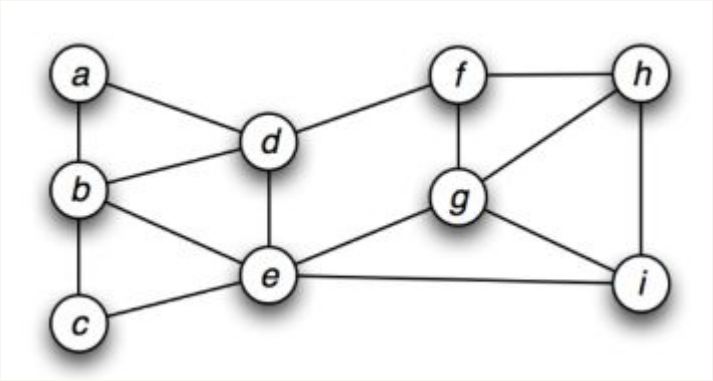
- Graph G can be **directed,** if E consists of ordered pairs, or **undirected**, if E consists of unordered pairs.
- If (u, v) Є E, then vertices u and v are adjacent.

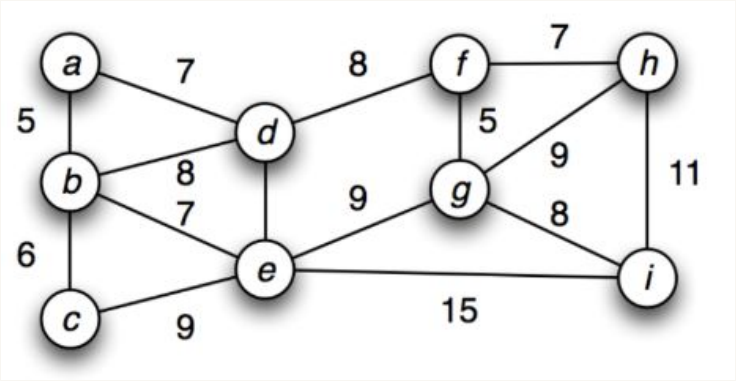




Weighted vs Unweighted Graphs

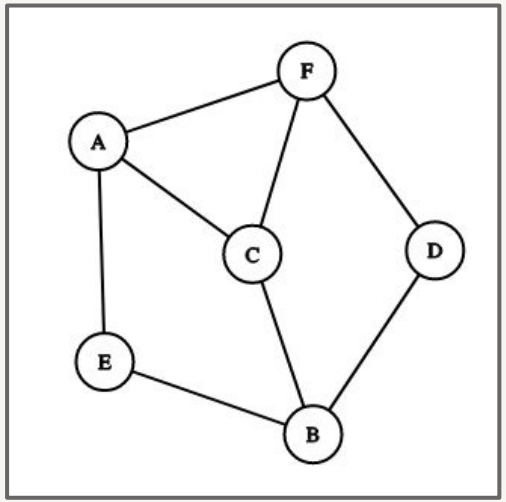
- Weights can be assigned to the edges i.e. **w_G (e) is a weight of the edge e Є E**. Any graph that has such function assigned to it's edges is called **weighted graph**.





Basic Terminologies

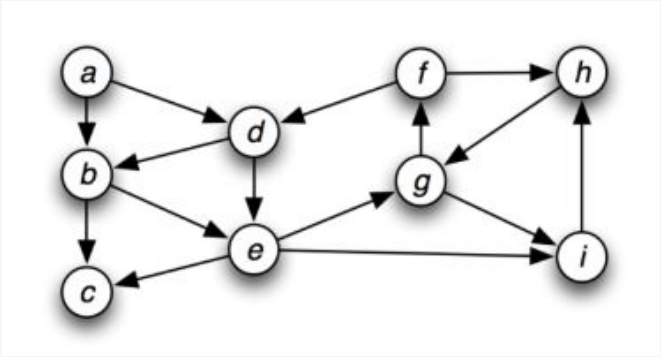
- **Edges:** Connections/Links between vertices (or nodes).



Here, AF, AC, BC etc. are edges of the graph.

Basic Terminologies

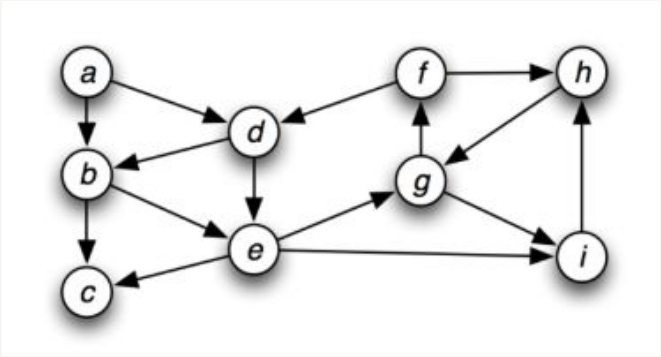
- **Incoming Edges:** Directed edges coming into a vertex.



Here, AB and DB are two incoming edges of vertex B.

Basic Terminologies

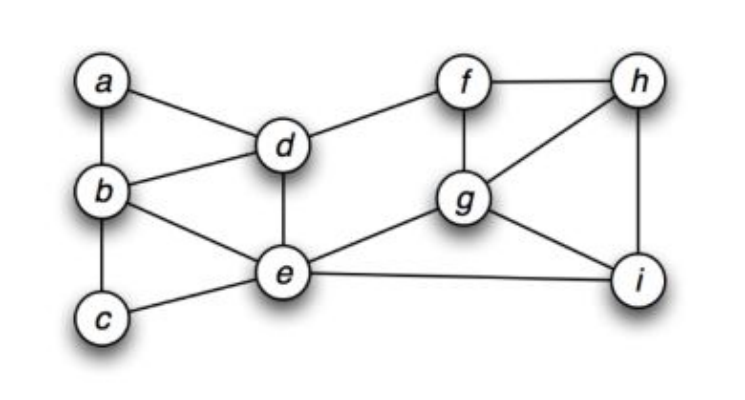
- **Outgoing Edges:** Directed edges going out of a vertex.



Here, BC and BE are two outgoing edges of vertex B.

Basic Terminologies

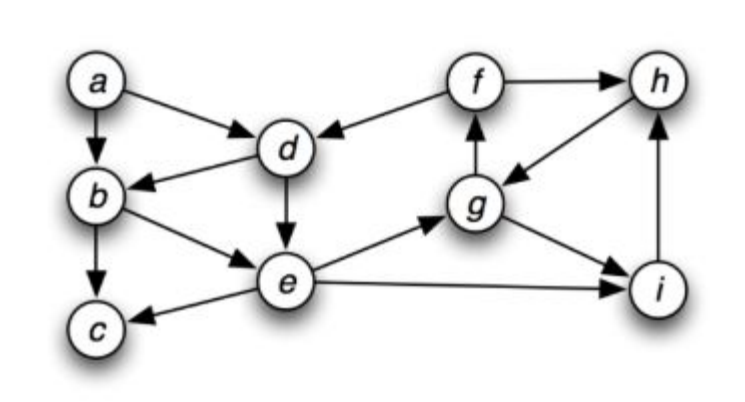
- **Degree:** For an undirected graph, **number of connected edges** to a vertex v is called the **degree of that vertex** and is denoted by **deg(v)**.



Vertex	deg(v)
a	2
b	4
c	2
d	4
e	5

Basic Terminologies

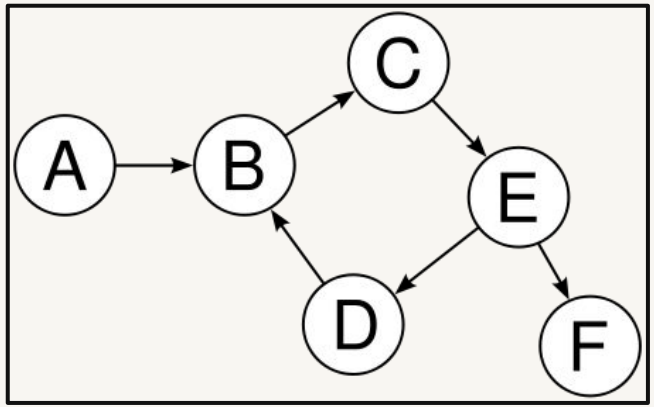
- **In Degree:** The **number of incoming edges** to a vertex v is called the **in-degree of the vertex** and is denoted by **indeg(v)**.
- **Out Degree:** The **number of outgoing edges** from a vertex v is called the **out-degree** and is denoted by **outdeg(v).**



Vertex	indeg(v)	outdeg(v)
a	0	2
b	2	2
c	2	0
d	2	2
e	2	3

Basic Terminologies

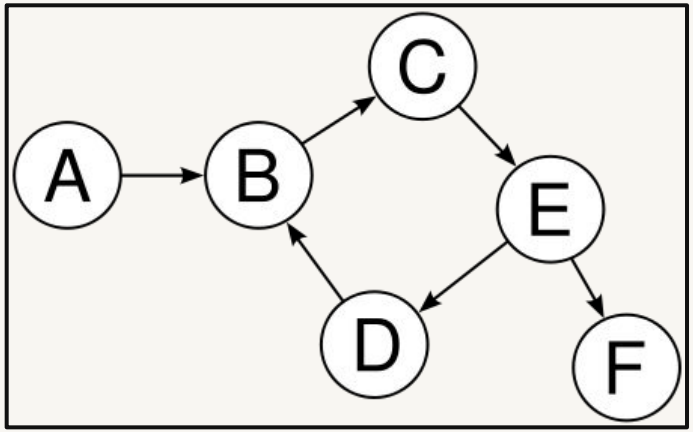
- **Source Vertex: A vertex with in-degree zero**. Generally, in a graph, **at least one** source vertex is present.



In the graph, vertex A is a source vertex.

Basic Terminologies

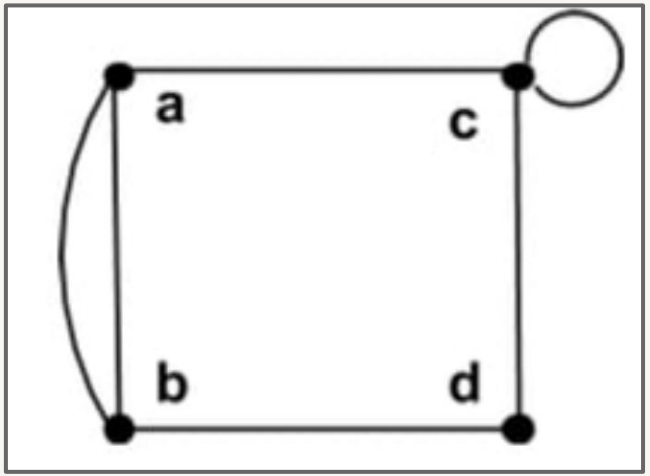
- **Sink Vertex: A vertex with out-degree zero.**



In the graph, vertex F is a sink vertex.

Basic Terminologies

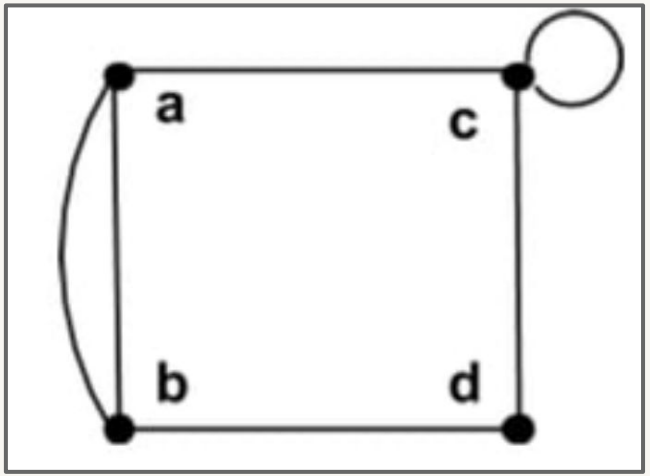
- **Parallel Edges: Multiple edges** between same pair of vertices.



In the graph, vertex a and b has two edges between each other i.e. they are parallel.

Basic Terminologies

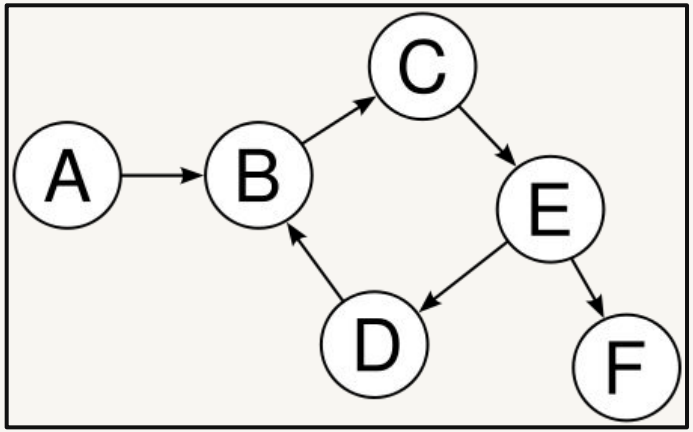
- **Self Loop:** An edge between **the vertex and itself**.



In the graph, vertex c has an edge which starts from c and ends at c i.e. a self loop.

Basic Terminologies

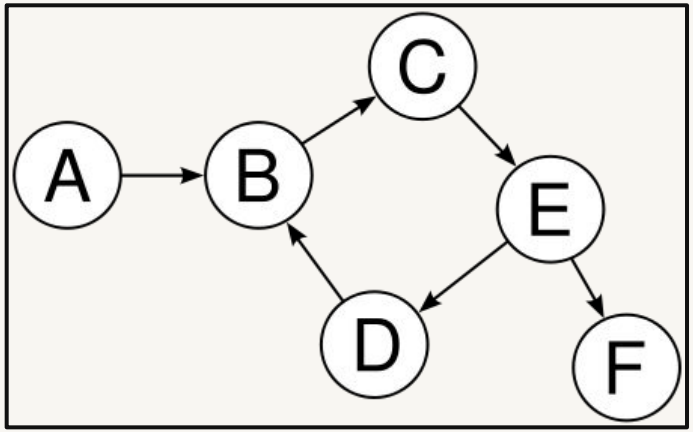
- **Path:** **A sequence of vertices connected by edge(s).**



In the graph, path from A to D is (A -> B -> C -> E -> D).

Basic Terminologies

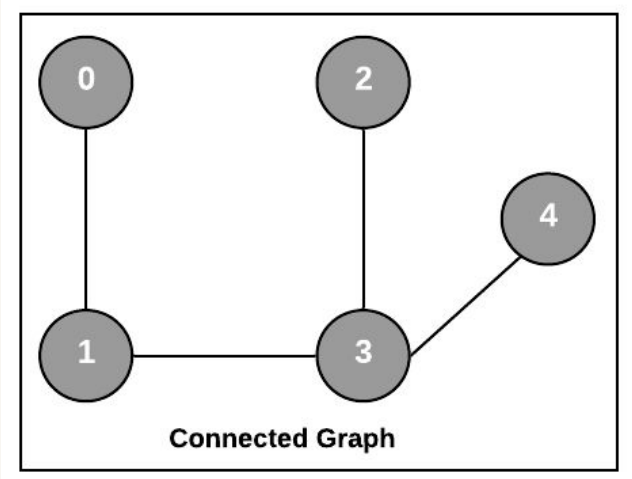
- **Cycle:** A closed path **where first and last vertices are the same**.



In the graph, a path exists from B to B (B -> C -> E -> D -> B) which is a cycle.

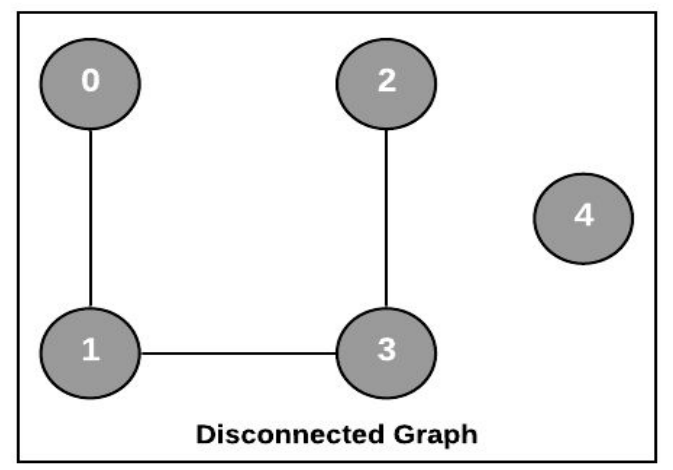
Basic Terminologies

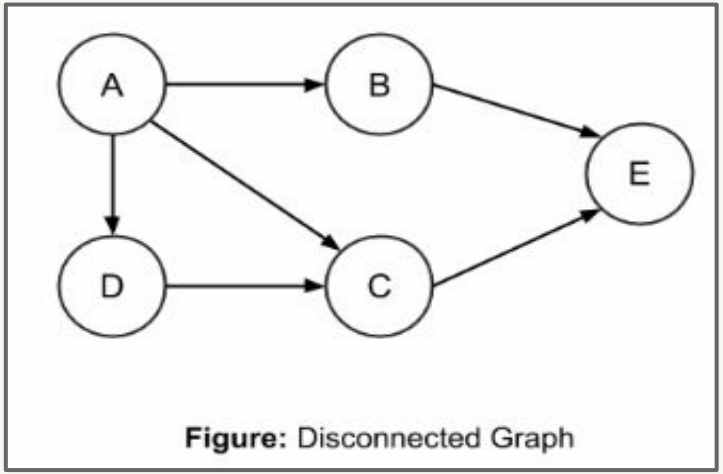
- **Connected Graph: A path exists between every pair of vertices.**



Basic Terminologies

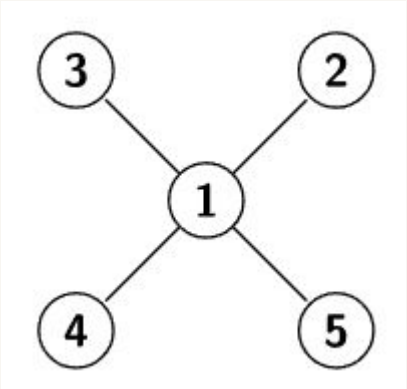
- **Disconnected Graph: A path is absent between any pair of vertices.**





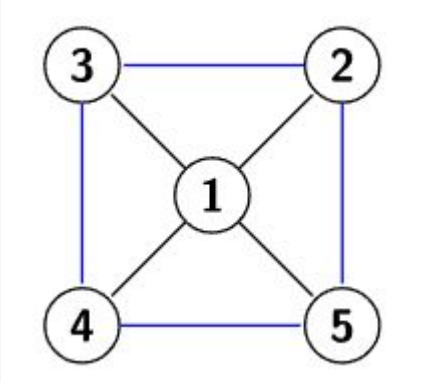
Basic Terminologies

- **Sparse Graph:** Number of edges is **considerably less** than maximum number of edges.



Basic Terminologies

- **Dense Graph:** Number of edges is **close to the maximum number** of edges.

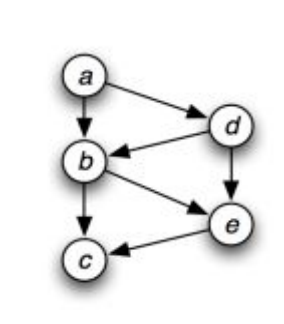


Relationship between n and m

- If G is a graph with m edges, then $\Sigma deg(v) = 2m$
- If G is a directed graph with m edges, then $\Sigma indeg(v) = \Sigma outdeg(v) = m$
- If G is a simple undirected graph with n vertices and m edges, then $m \leq n(n - 1)/2$.
- If G is a simple directed graph with n vertices and m edges, then $m \leq n(n - 1)$.

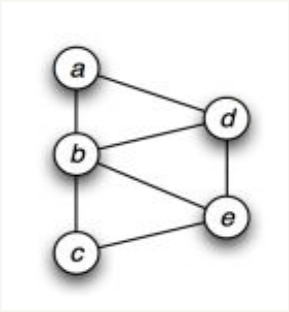
Representation of Graphs

- **Adjacency Matrix:**
  - A graph G = (V, E) can be represented in an matrix of size n x n.
  - If $(v_i, v_j) \in E$, then $a_{i,j} = 1$ and 0 otherwise. In case of undirected graph, $a_{i,j} = 1$ and also $a_{j,i} = 1$.
  - Memory complexity is **$O(n^2)$**



	a	b	c	d	e
a	0	1	0	1	0
b	0	0	1	0	1
c	0	0	0	0	0
d	0	1	0	0	1
e	0	0	1	0	0

Directed Graph

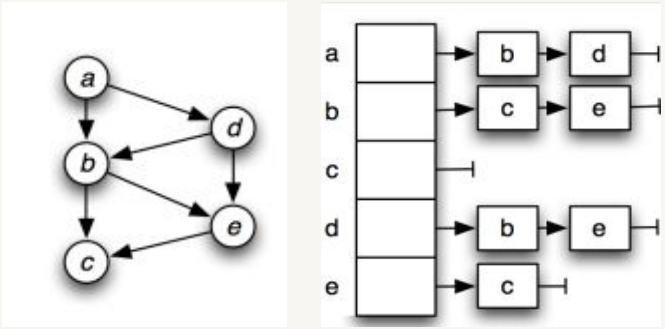


	a	b	c	d	e
a	0	1	0	1	0
b	1	0	1	1	1
c	0	1	0	0	1
d	1	1	0	0	1
e	0	1	1	1	0

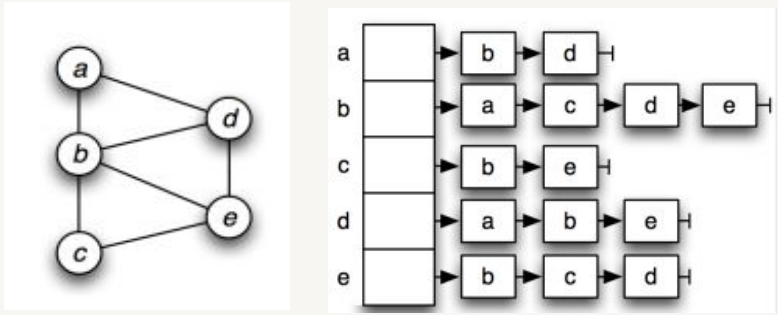
Undirected Graph

Representation of Graphs

- **Adjacency List:**
  - Represents the graph by listing for each vertex v_i, it's outgoing/connected vertices in a list ***out(v_i)***. (Can be represented using a Linked List or any other appropriate data structure)
  - Memory complexity is **O(m + n)**



Directed Graph



Undirected Graph

References

- Mumit sir’s slide - https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view

- PNP sir’s slide - https://docs.google.com/presentation/d/1bKKT_3Mnlp0cV509k7E8hBWEKQ_YfV0cfek10LDCo6Y/edit?slide=id.g26e6291b7d8_0_183#slide=id.g26e6291b7d8_0_183